

Uhlenbeck Integrals, a Rational Lax Matrix, and Compact Liouville Fibers for the Neumann Sphere

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Abstract

For the anisotropic Neumann oscillator on the round two-sphere we prove global completeness and derive all three Uhlenbeck integrals with their two linear relations. A rational 2×2 Lax matrix generates the integrals as determinant residues. Direct Dirac-bracket involution and compactness close the regular Liouville two-tori and their necessary-and-sufficient rational-frequency return criterion.

1 Frozen constrained system and theorem

Let $A = \text{diag}(a_1, a_2, a_3)$ with $a_1 < a_2 < a_3$. On

$$T^*S^2 = \{(x, p) \in \mathbb{R}^3 \times \mathbb{R}^3 : |x|^2 = 1, x \cdot p = 0\}$$

take

$$H = \frac{1}{2}(|p|^2 + x^T A x), \quad \dot{x} = p, \quad \dot{p} = -A x + \alpha x, \quad \alpha = x^T A x - |p|^2. \quad (1)$$

Put $L_{ij} = x_i p_j - x_j p_i$ and

$$F_i = x_i^2 + \sum_{j \neq i} \frac{L_{ij}^2}{a_i - a_j}. \quad (2)$$

Theorem 1. *The flow (1) is complete. The F_i are conserved, pairwise Dirac–Poisson commuting, and satisfy*

$$\sum_i F_i = 1, \quad \sum_i a_i F_i = 2H.$$

They are the residues of the rational Lax determinant in Section 3. Every connected regular common fiber of two independent integrals is a Liouville two-torus; the physical flow on it is periodic precisely when its frequency vector has a common period.

2 Completeness and the Uhlenbeck identities

The two constraint derivatives are

$$\frac{d}{dt}|x|^2 = 2x \cdot p, \quad \frac{d}{dt}(x \cdot p) = |p|^2 - x^T A x + \alpha = 0$$

on T^*S^2 . Also $\dot{H} = 0$. On $H = E$ the identity $|p|^2 = 2E - x^T A x$ bounds p ; the orbit remains in a compact subset, and the smooth vector field is complete.

The angular momenta satisfy

$$\dot{L}_{ij} = (a_i - a_j)x_i x_j. \quad (3)$$

Hence

$$\dot{F}_i = 2x_i \left(p_i + \sum_{j \neq i} L_{ij} x_j \right) = 0,$$

because the parenthesis equals $p_i + x_i(x \cdot p) - p_i|x|^2$. Pairing (i, j) and (j, i) proves $\sum F_i = 1$. The weighted pairing and the Gram identity give

$$\sum_i a_i F_i = x^T A x + \sum_{i < j} L_{ij}^2 = x^T A x + |p|^2 = 2H.$$

This is the *compact Uhlenbeck owner* added in revision round zero.

3 Resolvent Lax generator

For $\lambda \notin \{a_1, a_2, a_3\}$ let

$$R_\lambda = (\lambda I - A)^{-1}, \quad U = x^T R_\lambda x, \quad V = x^T R_\lambda p, \quad W = 1 + p^T R_\lambda p$$

and define

$$\mathcal{L} = \begin{pmatrix} V & U \\ -W & -V \end{pmatrix}, \quad \mathcal{M} = \begin{pmatrix} 0 & 1 \\ \alpha - \lambda & 0 \end{pmatrix}. \quad (4)$$

Using $a_i/(\lambda - a_i) = \lambda/(\lambda - a_i) - 1$ in (1) gives

$$\dot{U} = 2V, \quad \dot{V} = W + (\alpha - \lambda)U, \quad \dot{W} = 2(\alpha - \lambda)V.$$

These are exactly $\dot{\mathcal{L}} = [\mathcal{L}, \mathcal{M}]$. Moreover,

$$\det \mathcal{L} = UW - V^2 = \sum_{i=1}^3 \frac{F_i}{\lambda - a_i}. \quad (5)$$

Indeed, the residue at a_i is (2); both rational functions vanish at infinity and have no other poles.

The constrained bracket is

$$\{x_i, x_j\}_D = 0, \quad \{x_i, p_j\}_D = \delta_{ij} - x_i x_j, \quad \{p_i, p_j\}_D = x_j p_i - x_i p_j.$$

Differentiating (2), substituting these three coordinate brackets, and pairing the denominators $a_i - a_j$ and $a_j - a_i$ gives

$$\{F_i, F_j\}_D = 0 \quad (1 \leq i, j \leq 3). \quad (6)$$

The full finite-sum cancellation is recorded in the proof package and is independently expanded by the symbolic lane.

The two linear relations leave two generically independent integrals. A regular common fiber is closed in a compact energy shell. Liouville–Arnold therefore identifies each connected component with $\mathbb{T}^2$ and gives $\theta(t) = \theta(0) + t\omega$. Return occurs exactly when

$$T\omega \in 2\pi\mathbb{Z}^2 \quad \text{for some } T > 0. \quad (7)$$

For two nonzero components this is $\omega_1/\omega_2 \in \mathbb{Q}$. Continuously selected resonant tori are not an isolated primitive-orbit ledger. Equations (4)–(7) are the *resolvent torus owner* of revision round one.

Source boundary

Neumann owns the original mechanical problem; Moser owns the modern quadrics/spectral lineage; Knoerrer owns the geodesic correspondence cited here. This convention-locked reconstruction makes no priority claim.

References

- [1] C. Neumann, “De problemate quodam mechanico, quod ad primam integralium ultraellipticorum classem revocatur,” *J. Reine Angew. Math.* 56 (1859), 46–63, DOI 10.1515/crll.1859.56.46.
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- [3] H. Knoerrer, “Geodesics on quadrics and a mechanical problem of C. Neumann,” *J. Reine Angew. Math.* 334 (1982), 69–78, DOI 10.1515/crll.1982.334.69.